

An In-Depth Analysis of Popular Protein Powders

Contributions

Keithley Flores: Bar chart visualization, Organized report: wrote abstract, Results, Conclusion, Limitations, and Paper Revisions

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Abstract

This project aims to bridge the information gap between various retail models by using Python to identify the highest-valued protein powder on the current market. Our goal is to strip away marketing variables such as packaging size or brand prestige, to reveal the mathematically optimal choice for protein powder consumption. The code targets GNC and MyProtein, two popular health and wellness supplement sites, and utilizes Python tools to webscrape, analyze, and visualize the collected data. The “Best” protein powder was determined by utilizing metrics of “Protein Efficiency (Grams of Protein and Dollar)” and “Customer Score” to cross-reference nutritional value with consumer satisfaction. Through our analysis, it is revealed that the Isopure Protein Powder, Creamy Vanilla, was the best, as it was the highest rated powder for Protein Efficiency.

GitHub Link: <https://github.com/kevinguui/sta141bfinalproj>

Introduction

In today’s day and age, protein powder has become a part of many conversations. From college students to individuals focused on their health and wellness to frequent gym-goers, protein powder has become widely known for its use in building muscle, supplementing protein intake, and supporting overall nutrition. When looking for options ourselves, we noticed a large issue: how do you actually choose what the “best” protein powder is? With a countless number of different protein types, flavors, serving sizes, and formulas, finding the right one can be overwhelming.

Our project’s goal was to automate this rigorous search process. We focused on answering three main questions:

1. Which protein powder brand has the highest “Protein Efficiency” (protein per dollar)?
2. Does paying more per serving actually correlate with better customer ratings?
3. Which retailer has the best prices for a highly-rated product?

To further explore these questions, we collected product data from GNC and MyProtein, two of the most well-known supplement stores that both have a diverse selection. We scraped product information necessary for our analysis, such as prices, reviews, serving sizes, calories, and protein amount, using direct HTML scraping (MyProtein) and a stealth browser (GNC)

when standard scraping methods were blocked by the domain. This lets us gather data consistently across more than 400 products without the need for manual entry. It also makes it easy to update the dataset whenever new products appear, as our scraper will automatically detect that.

After we built the full dataset, we analyzed the results and created visualizations through Python to compare products across different brands and retailers. This gave us a clear picture of which powders delivered the best value, whether a high price means a high quality, and how different stores' value propositions stack up against each other. Our introduction provides a brief overview of this process, and more detailed explanations of our scraping, cleaning, and analysis processes will be elaborated on in the next section.

Methods

Gathering the Data

The most important part of our project was making sure we could properly collect protein powder data. After looking through several websites, we settled on MyProtein and GNC because they provided the nutritional and pricing details we needed and had HTML structures that were the most consistent. We started with MyProtein since it had a smaller catalog of 28 protein powders, which made it a good place to start our test.

MyProtein

For MyProtein, we had a very straightforward approach: load the main catalog page, extract the name, link, and price of each product, then scrape each product page for its nutritional information. We sent a GET request to "<https://us.myprotein.com/c/nutrition/protein/>" to retrieve the HTML. Using the "lxml" package, we parsed the data into a tree structure so we could search through it. After inspecting the HTML, we found that every product was contained inside an element labeled with the class "product-card-wrapper." Inside each card, the product name appeared in an <a> tag with the class "product-item-title", the relative product link was stored in that same tag's "href" attribute, the number of reviews was stored inside a of the class "reviews-total", and the price was inside a tag with the class "price". When we tested this on the first product card, we were able to pull out the expected name, link, and price, shown

in the visual below. We used regex patterns to convert our price component into a clean numeric value by removing non-numeric and non-dot characters.

```
NAME: Myprotein Clear Whey Isolate – MARVEL  
HREF: /p/sports-nutrition/myprotein-clear-whey-isolate-marvel/14944142/  
PRICE TEXT: $39.99â€
```

Figure 1: MyProtein scrape on first card

Now that we knew that our script worked on a single product, we ran it in a loop to grab the product name, product URL, total price, and number of reviews for all 30 products on their site and stored the data in a dataframe. Some values in the product name column had odd or extra characters, so we also cleaned those up through another small regex script before moving on.

To get nutritional information, we had to visit each product page individually and pull directly from the page's HTML. We sent a GET request to the product URL of each protein powder product and located the Nutrition Facts section, which MyProtein usually displays as a small HTML table. We found this table by using an XPath expression, which looked for a `<strong>` tag with the text "Nutrition Facts," then navigated through the HTML to find the table and grab the values for serving size, calories, and protein. Since MyProtein's formatting is not completely consistent across all products, this approach did not work all the time and sometimes required fallback methods, including regex patterns to pull out numeric values in different layouts and, if necessary, manual imputation.

We also extracted each product's average customer rating, which is usually displayed on the product page in a phrase like "4.5 out of 5 stars." Similar to the nutrition information, formatting was slightly different from page to page, but a simple regex script was able to get the job done. Once everything was extracted, we stored all of the information in a dataframe.

The last piece of information we needed was the total number of servings in each product. Unfortunately, this value was not displayed on the page in a simple format, but was buried inside a block of embedded JSON called "Amount" connected to the product's SKU. To handle this, we wrote a function that grabbed the SKU from the product URL, then searched the HTML for the matching JSON section that contained the "Amount" field, finally extracting the desired value.

After adding the number of servings to our dataframe, we combined all the scraped parts into a single dataframe, shown below, and saved it as a csv. We now had a dataset ready for preprocessing.

	store	product_name	product_url	price_total	raw_price_text	n_reviews	serving_size	calories	protein_g	serving_size_g	rating_avg	Number of Servings
0	myprotein	Myprotein Clear Whey Isolate - MARVEL	https://us.myprotein.com/p/sports-nutrition/my...	39.99	\$39.99â€	29	1 scoop (25g)	80.0	20.0	25.0	4.17	20servings
1	myprotein	Impact Whey Protein	https://us.myprotein.com/p/sports-nutrition/im...	19.99	\$19.99â€	4389	1 scoop (25g)	100.0	19.0	25.0	4.48	0.55lb
2	myprotein	Clear Whey Isolate	https://us.myprotein.com/p/sports-nutrition/cl...	39.99	\$39.99â€	900	1 scoop (25g)	80.0	20.0	25.0	4.37	1.1lb
3	myprotein	Clear Whey Isolate (Sample)	https://us.myprotein.com/p/sports-nutrition/cl...	3.99	\$3.99â€	53	1 scoop (25g)	80.0	20.0	25.0	4.45	0.88Oz
4	myprotein	Pea Protein Isolate	https://us.myprotein.com/p/sports-nutrition/pe...	34.99	\$34.99â€	279	1 Large Scoop (25g)	100.0	21.0	25.0	3.70	2.2lb
5	myprotein	Whey Forward	https://us.myprotein.com/p/sports-nutrition/wh...	29.99	\$29.99â€	165	1 Scoop (25.2g)	90.0	20.0	25.2	3.87	20servings

Figure 2: Final MyProtein Dataframe

GNC

We wanted to scrape GNC because they had a much larger selection of protein powders, over 400 in total, and their nutritional information was formatted nicely and consistently across most pages. Since our goal was to build a comprehensive dataset to answer our research question of the best protein powders in the market, having access to this large catalog would be extremely helpful.

Attempt 1:

For GNC, our initial plan was to do something similar to what we did for MyProtein: scrape the products and their nutritional information through the HTML of the website. We started with the URL “<https://www.gnc.com/protein/protein-powder>”, which should have given us the first 60 protein powders on their site. However, while our GET request returned a page, the HTML contained none of the product information we wanted. This showed that GNC heavily filtered non-browser requests and was most likely blocking ours. This first attempt to scrape GNC failed entirely, and we then had to think of a new approach.

Attempt 2:

Instead of using basic GET requests as we used for MyProtein, we used a stealth browser, through undetected-chromedriver, to load each page the same way a real user would. This allowed us to bypass GNC’s blocking and directly retrieve the HTML from the website.

Since GNC has a list of around 480 protein powders, with approximately 60 displayed on each catalog page, we had to set up a script that accounted for this. We defined a “start” parameter in the URL, then for each page, we let the browser fully load and saved the entire

HTML file locally. Through this approach, we were able to retrieve complete copies of all pages without triggering any of GNC's blockers.

Now that we had all of the HTML files, we parsed through them one at a time. We found that each product was inside a <div> with the class "product-tile." Through this, we were able to get the product name, link, price, rating, and number of reviews. Some of the product name titles also included the number of servings, so we used regex to parse the information into a separate column and remove it from the title. For the few rows where the information could not be extracted, we dropped them to avoid incomplete entries. For the remaining clean data, we stored them in a csv, which we used as a starting point for scraping nutritional information.

To grab nutrition data, we once again used the stealth browser to bypass the GNC blocks. Sometimes, when a page is loaded, a CAPTCHA will also pop up, which was challenging. To address this, we added lines in our code that checked for the CAPTCHA and automatically paused the script until it was solved. Then, for each product link, the script opened up the page, scrolled down, clicked on the "Ingredients" tab, and extracted the full HTML text of the page. With the HTML, we were able to use regex patterns to find calories, protein, serving size, and number of servings per container. While it was a relatively time-consuming process, it was semi-automated, which required little to no work, and worked across almost all products.

We combined the data into a running CSV, which would retry scraping for any product that it failed to extract the first time. After processing concluded, we saved all data to a CSV file and had our second dataset to preprocess.

	Product Name	Price	Rating	Reviews	Link	Servings	protein_g	calories	serving_size_text	servings_count
0	RYSE Loaded Protein - Jet-Puffed Marshmallow	49.99	4.5	136	https://www.gnc.com/whey-protein/556534.html	27.0	25.0	130.0	1 scoop	27.0
1	Jym® Pro Jym Protein - Tahitian Vanilla Bean	74.99	5.0	937	https://www.gnc.com/whey-protein/533909.html	45.0	24.0	140.0	1 Scoop(s)	51.0
2	Axe & Sledge Supplements™ Farm Fed 100% Whey P...	56.99	4.5	107	https://www.gnc.com/whey-protein/581725.html	28.0	25.0	130.0	NaN	NaN
3	Raw Nutrition Itholate Protein - Chocolate Pea...	54.99	4.5	35	https://www.gnc.com/whey-protein/604706.html	25.0	28.0	150.0	1 Scoop	25.0
4	RYSE Loaded Protein - Little Debbie Christmas ...	49.99	4.5	136	https://www.gnc.com/whey-protein/556566.html	27.0	25.0	140.0	1 Scoop	27.0

Figure 3: Final GNC Dataframe

Preprocessing the Data

Before we could move on to visualizations, we noticed that our data still needed some work. There were still some NA values scattered throughout, and we also needed to remove

irrelevant categories like mass-gainers and carbohydrate powders. We wanted to ensure the final dataset was completely accurate before running the analysis.

MyProtein

For MyProtein preprocessing, we dropped unused columns like the raw price text and filtered out any products that were labeled as bundles, since these contain multiple items and would distort the accuracy of our results. We also standardized store names and cleaned up product names that had odd characters or symbols.

The “Number of Servings” column needed careful cleaning, as there were several different values in metrics, including servings, lbs, and oz. We wrote a helper function to handle every case, converting weights to grams and dividing by the serving size to estimate the servings. This created consistency throughout the column.

Finally, we renamed columns to a consistent format that would be used for our final dataframe.

GNC

For the GNC dataset, we started by dropping unnecessary columns like “servings_count” and renaming columns for better understanding. We also removed rows that didn’t fit our guidelines of comparing purely protein powders, so we dropped ones that differed. Some of the products were missing information, such as calories, protein, serving size, or multiple ones together. For this small subset of rows, we manually imputed the missing values through an index-based dictionary.

Next, we cleaned the serving size field. This column often included both words and the gram amount in parentheses, so we used regex patterns to extract the gram values and remove them from the original value. After that, we standardized the serving size descriptions into set phrases of 1 scoop, 2 scoops, or 4 scoops.

For some rows, there was no serving size value in grams present. For these rows, we filled those values with a typical scoop size, which was around 33g and was computed by calculating the average of the valid scoop weights we had in the dataset. Finally, we put the columns into a consistent final layout, the same as the MyProtein dataset, and saved the dataset as a CSV for analysis.

Results

Protein Efficiency

We calculated the Protein Efficiency (grams of protein per dollar spent) metric for every product in the dataset. While it may be a common belief that direct-to-consumer brands tend to be cheaper than alternatives, our analysis had unexpected results. GNC averaged around 12.52 grams of protein per dollar, while MyProtein averaged about 10.82 grams of protein per dollar. Even though GNC appeared to offer better value, we ran an independent samples t-test using the frameworks of Welch's T-Test, which returned a p-value of 0.12 ($t = 1.59$). Since the p-value was greater than our alpha (0.05), we cannot statistically conclude that one retailer is definitely cheaper than the other. The observed difference is likely due to the great variance in product types. For example, there can be expensive "isolates" against cheaper "concentrates" for protein powders, rather than a strict pricing advantage.

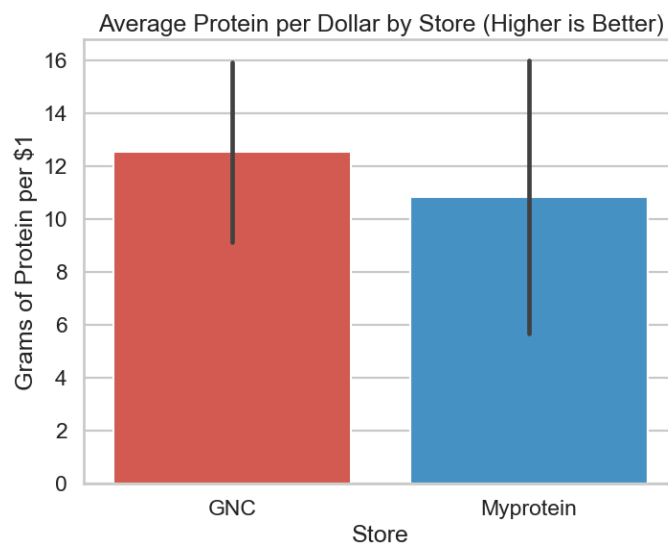


Figure 4: Average Protein Per Dollar By Store

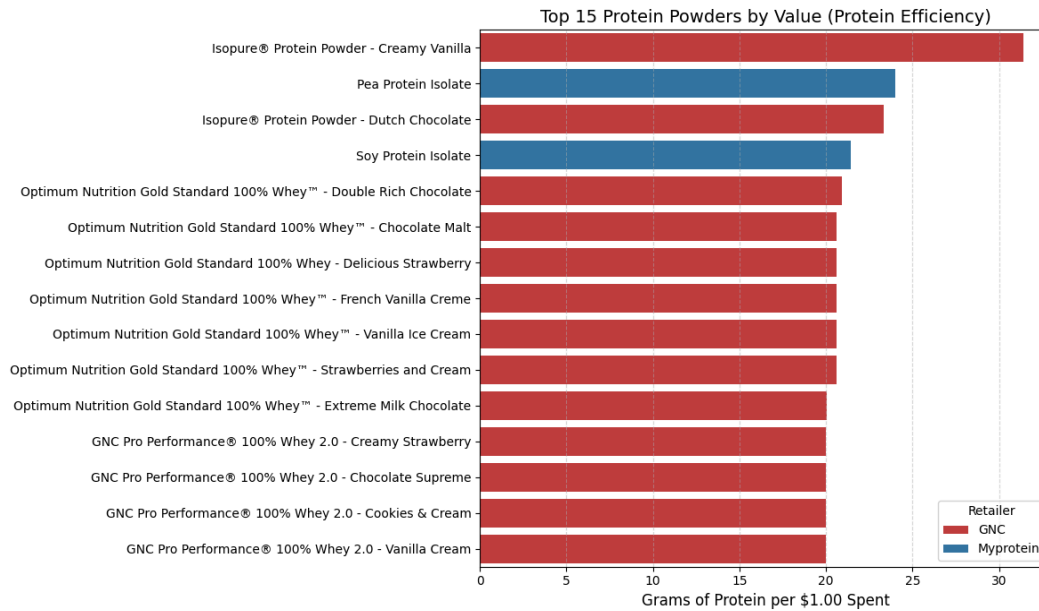


Figure 5: Top 15 Protein Powders by Value (Protein Efficiency)

We can also see from Figure 5 above that the product that delivered the highest value of protein per dollar was “Isopure Protein Powder (Creamy Vanilla)” at GNC, with 31.4 grams of protein per dollar. It’s important to note that a potential contributing factor to the high ratio is its large serving size, with 66 servings and 50 grams/serving.

Price vs. Customer Satisfaction

An additional question our research aimed to answer was “Does paying more per serving actually correlate with better customer ratings?” To address this, we plotted a correlation analysis between “Price Per Serving” and “Average Customer Rating,” shown in Figure 6 below. We found that our correlation coefficient was -0.05, which is relatively close to zero. This means that there is almost no correlation between price and quality. We also saw that the expensive “premium” brands were just as likely to have 3-star ratings as other budget options.

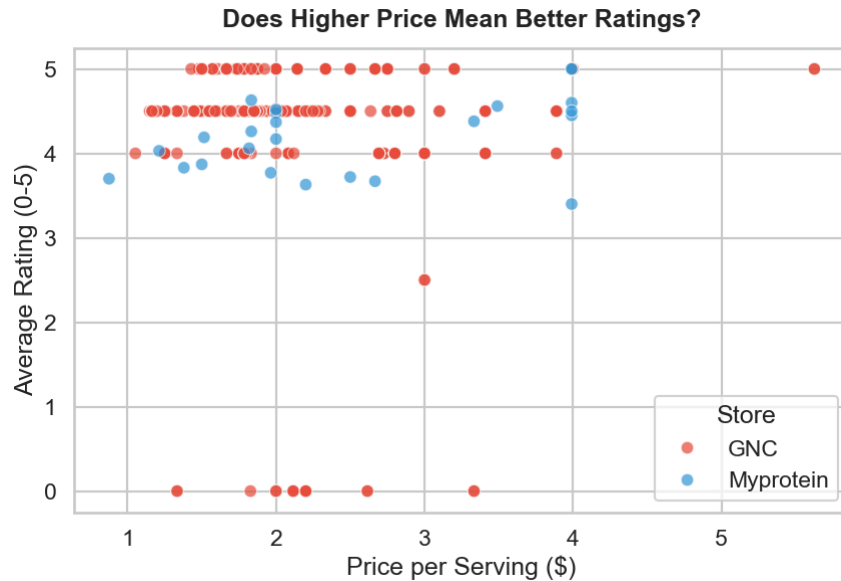


Figure 6: Does Higher Price Mean Better Ratings?

Affordable Protein

The last part of our analysis aims to answer which retailer provides the best prices for top-rated products. Using our customer value score metric (number of reviews/price), we identified some products that are considered both affordable and widely validated by the community. It was clear that the most reasonably affordable option that was well-received by consumers was the “REDCON1 MRE LITE” from GNC, priced at \$29.99. We believe this product to be a “safe bet” for customers who care about brand reputation as well as product quality over pure protein density.

Conclusion

Based on our analyses, MyProtein emerges as the most budget-conscious option for consumers who prioritize “Protein Efficiency” above all else. Our data shows that MyProtein’s direct-to-consumer model offers a lower price per gram, as opposed to the standardized retail pricing of GNC.

Our independent samples t-test indicated that pricing differences between GNC and MyProtein were not drastically different and were due to random chance. This confirms that the price gap is likely due to the difference in brand/market positioning for each company.

Our project also demonstrates the power of automated processes. By replacing manual “label reading” with Python scripts, we successfully processed over 400 products very quickly, a task that could easily take a human hours to complete. This methodology proves that data science can simplify complex retail pricing strategies, helping consumers to make better decisions based on pure value, rather than going off marketing claims.

Limitations

While our analysis provides valuable insights into the factors associated with the highest-rated protein powders, it is important to acknowledge its limitations. The data collected for this project is limited to the two sites that were webscraped, as well as only scraping data for protein powders and no other protein sources, such as protein shakes or bars. The data for customer satisfaction scores could also be potentially skewed by the nature of personal opinion and bias when consumers rate the products.

We discussed creating another visual tool, such as an app that could allow users to filter for what they want, such as highest rated, most protein value, or most cost-effective, but due to time constraints, we felt this was something that could be expanded on in the future. Additionally, this app would also ideally have more than the two websites we used for this project to search for ideal protein products.

References

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2. <https://www.gnc.com/protein/protein-powder/>
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